

# Lecture 1: Introduction to Generalised Linear Models

## MATH3701 Statistical Modelling

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**Course notes:** Chapter 1 (Introduction)

MATH3701 Statistical Modelling notes, Chapter 1

## What we will cover:

- ➊ Revision of linear models with normal errors
- ➋ Introduction to generalised linear models (GLMs)
- ➌ Logistic regression models
- ➍ Loglinear models and contingency tables

**Key idea:** GLMs extend normal linear models to handle responses that are *not necessarily normal*.

**Prerequisites:** Comfort with R programming and basic statistical computation.

# The Fundamental Question

**Goal:** Describe how a response variable  $Y$  depends on  $p$  explanatory variables  $\mathbf{x} = (x_1, x_2, \dots, x_p)$ .

**Normal linear model assumption:**

$$Y \mid \mathbf{x} \sim \mathcal{N}(\mu(\mathbf{x}), \sigma^2)$$

**The problem:** Many real-world responses are:

- Binary (success/failure, yes/no)
- Counts (number of events)
- Proportions (bounded between 0 and 1)
- Positive continuous (times, costs)

$\Rightarrow$  The normal distribution is often inappropriate.

## Motivating Example: Beetle Mortality

**Dose-response experiment:** Beetles exposed to carbon disulphide gas.

| Dose ( $x_i$ ) | Beetles ( $m_i$ ) | Killed ( $y_i$ ) |
|----------------|-------------------|------------------|
| 1.6907         | 59                | 6                |
| 1.7242         | 60                | 13               |
| 1.7552         | 62                | 18               |
| 1.7842         | 56                | 28               |
| 1.8113         | 63                | 52               |
| 1.8369         | 59                | 53               |
| 1.8610         | 62                | 61               |
| 1.8839         | 60                | 60               |

**Question:** How does mortality rate depend on dose?

# Why Linear Regression Fails

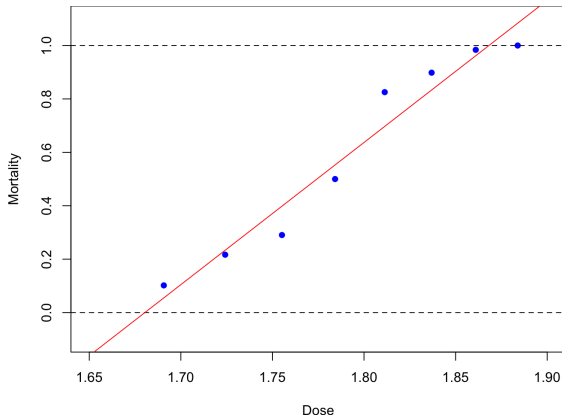
## Fitted linear model:

$$\hat{p} = \hat{\alpha} + \hat{\beta}x$$

where  $p = y/m$  is the mortality proportion.

## Problems:

- Predicts  $p > 1$  at high doses
- Predicts  $p < 0$  at low doses
- Mortality is *bounded*  $[0, 1]$



# A Better Approach: The Logistic Model

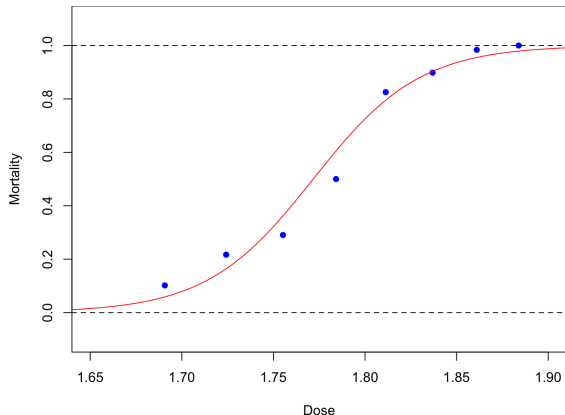
Logistic function:

$$p = \frac{e^{\alpha + \beta x}}{1 + e^{\alpha + \beta x}}$$

Properties:

- S-shaped (sigmoid) curve
- Always between 0 and 1
- Fitted via maximum likelihood

This is an example of a **Generalised Linear Model**.



## Revision: The Linear Model

For  $n$  paired observations  $(x_i, y_i)$ :

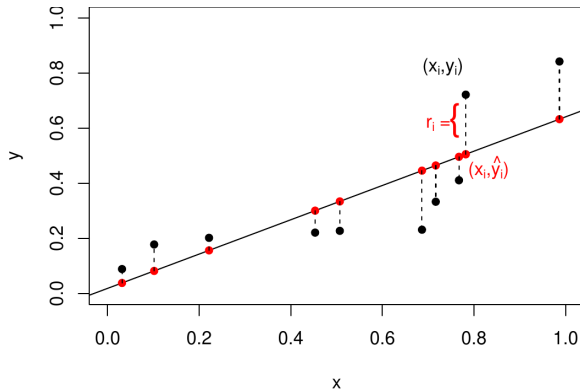
$$y_i = \alpha + \beta x_i + \epsilon_i$$

### Assumptions:

- Errors  $\epsilon_i$  are independent
- $\mathbb{E}[\epsilon_i] = 0$
- $\text{Var}[\epsilon_i] = \sigma^2$  (constant variance)

**Goal:** Estimate  $\alpha$ ,  $\beta$ , and  $\sigma^2$  from data.

# Visualizing the Fitting Problem



The data points are fixed; we choose  $\alpha, \beta$  to move and rotate the line. Each point has a vertical residual  $r_i = y_i - (\alpha + \beta x_i)$ ; RSS sums the squared residuals.

# Least Squares Estimation

## Residual Sum of Squares:

$$\text{RSS}(\alpha, \beta) = \sum_{i=1}^n (y_i - (\alpha + \beta x_i))^2$$

Minimize RSS to obtain the **least squares estimators**:

$$\hat{\beta} = \frac{s_{xy}}{s_x^2}, \quad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

where:

- $\bar{x}, \bar{y}$  are sample means
- $s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$  is sample covariance
- $s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$  is sample variance of  $x$

# Properties of Least Squares Estimators

## Unbiasedness:

$$\mathbb{E}[\hat{\alpha}] = \alpha, \quad \mathbb{E}[\hat{\beta}] = \beta$$

## Fitted values and residuals:

- Fitted values:  $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$
- Residuals:  $r_i = y_i - \hat{y}_i$

## Error variance estimation:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n r_i^2$$

This is an unbiased estimator:  $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$ .

# Types of Variables

## Quantitative variables:

- **Continuous:** Real-valued measurements (height, weight, time)
- **Count (discrete):** Non-negative integers (number of events)

## Qualitative (categorical) variables:

- **Ordinal:** Ordered categories (mild/moderate/severe)
- **Nominal:** Unordered categories
  - **Binary/Dichotomous:** Two categories (yes/no, male/female)
  - **Polytomous:** Multiple categories (blood type, eye color)

⇒ Variable type determines the appropriate modeling approach.

# Looking Ahead: The GLM Framework

## Three components of a GLM:

- ① **Random component:** Distribution of  $Y$  (exponential family)

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}$$

- ② **Systematic component:** Linear predictor

$$\eta = \sum_{j=1}^p \beta_j x_j = \mathbf{x}'\boldsymbol{\beta}$$

- ③ **Link function:** Connects mean to linear predictor

$$\eta = g(\mu), \quad \mu = g^{-1}(\eta)$$

Normal linear regression is a special case with  $g(\mu) = \mu$  (identity link).

## Key points from today:

- Linear regression assumes normal errors, which is often inappropriate
- GLMs extend linear models to non-normal responses
- The beetle example shows why we need bounded response models
- Least squares estimation minimizes RSS
- Variable classification guides model choice

**Next lecture:** Normal linear models in detail — matrix formulation and model fitting in R.